

*Zeal Education Society's***ZEAL COLLEGE OF ENGINEERING & RESEARCH, PUNE-41**

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

NBA Accredited, NAAC Accredited with A+ Grade / ISO 21001:2018

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING**PROJECT BASED LEARNING****PROJECT SYNOPSIS****1. PROJECT TITLE****CampusShare** – Student Resource Sharing & Mutual Exchange Platform**2. TEAM MEMBERS**

Sr. No.	Name of Student	Roll No.	ZPRN
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2	[Partner Name 1]	[Roll No 1]	[ZPRN 1]
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3. DOMAIN

Web Application Development, Secure Database Systems & Communication APIs

4. INTRODUCTION

Academic resources represent a recurring financial burden for college students. These resources are often only needed for a single academic term, after which they remain idle. **CampusShare** is a secure peer-to-peer web platform designed specifically for students to catalog, rent, barter, or donate academic items. The application restricts membership to verified campus students by utilizing an automated registration flow. Students register by providing their unique ZPRN number, and the backend system automatically cross-references it with the ECE department's student directory to lookup their email address and deliver a 6-digit activation code via Gmail SMTP.

5. PROBLEM STATEMENT

College students face significant expenses purchasing study materials and gear that they only use temporarily. While informal peer sharing exists, it lacks coordination, visibility, and accountability. Current social media

groups are disorganized, prone to spam, and provide no protection against lost, damaged, or unreturned items. There is a critical need for a structured campus web portal that restricts access strictly to verified ECE department students, automates verification to protect user security, and provides a rating and Trust Score framework to hold students accountable for the items they borrow and return.

6. OBJECTIVES

- **ZPRN-Based Security:** Create an automated student registration validation that maps ZPRN inputs to database emails and emails verification codes.
- **Resource Catalog:** Build clean listings interfaces for students to catalog study guides, instruments, or electronic kits for Rent, Sale, or Exchange.
- **Transaction Tracker:** Model operational states (Request -> Approve -> Handover -> Return) to monitor items.
- **Trust & Score Metrics:** Implement a dynamic peer rating and Trust Score system to reward reliable members and moderate bad actors.
- **Interactive Dashboard:** Build a modern, mobile-responsive layout for student activity feeds, lost-and-found reports, and message threads.

7. LITERATURE SURVEY

Manual WhatsApp/Telegram Groups: Students coordinate sharing through chat groups. *Limitation:* Messages are easily buried, listings are unorganized, and there is no verification system to prevent non-students from joining.

Commercial Rental Websites: *Limitation:* Heavy transaction fees, no focus on student-to-student sharing, and lacks trust mechanisms suited for a campus community.

Static College Web Directories: Some colleges host static resource pages. *Limitation:* Static text files cannot handle interactive student bookings, transaction tracking, or user accountability.

8. PROPOSED SYSTEM

The proposed **CampusShare** platform is structured around a Node.js Express backend and a responsive Tailwind CSS frontend.

- **ZPRN-Verified Registration:** Users register using their ZPRN. The backend resolves their registered email and sends an activation code using Gmail SMTP, eliminating manual email input errors.
- **Mutual Sharing Models:** Support for Listings, Renting (dates and deposits), and Exchange (swapping items).
- **Trust Score & Moderation:** Calculates a rolling trust index for each student profile based on borrow history, return delays, and peer ratings.
- **Lost & Found Channel:** Integrated portal for students to catalog lost or found items on campus, complete with description fields and contact tabs.
- **Responsive Workspace:** Clean dashboard featuring user statistics, top shared items, and active metrics.

9. SYSTEM ARCHITECTURE

The system utilizes a 3-tier architecture: the frontend browser interface, the backend Node.js server, and local database stores. The user enters their ZPRN number on the web client, which sends an API request to the backend. The backend queries the ECE database directory, retrieves the email, dispatches a verification code through Gmail SMTP, and registers the activated account locally.

10. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

Sr. No.	Hardware	Specification
1	Development PC	Intel i3/i5 or AMD Ryzen 3/5 Processor, 8GB RAM
2	Storage	SSD with minimum 10GB free space
3	Network	Internet connection (for SMTP email dispatch)

Software Requirements:

Sr. No.	Software / Technology	Version / Details
1	Operating System	Windows 10 / 11 (64-bit)
2	Code Editor	Visual Studio Code (VS Code) (with AI coding assistance)
3	Runtime Environment	Node.js v18.0+ & npm
4	Local Testing Web Server	Apache Server (XAMPP for PHP script verification)
5	Version Control	Git (Portable Edition v2.45+)

11. TECHNOLOGIES / TOOLS USED

- **Frontend:** HTML5, Tailwind CSS, Lucide Icon sets
- **Backend API Server:** Node.js, Express Framework, CORS middleware
- **Email Delivery Service:** Nodemailer library integrated with secure Google Gmail SMTP
- **Local Databases:** Node JSON file-based database, LocalStorage API
- **Development Tools:** Visual Studio Code (VS Code) + AI Coding Assistant, Git (Version Control)

12. EXPECTED OUTCOME

The project will deliver a fully-functioning, secure campus resource sharing web application. ECE students will be able to register using their ZPRN numbers, verify their email accounts, upload resources, request items from peers, and track active transactions. This platform will reduce student expenses, minimize resource waste, and foster a trusted environment on campus.

13. SCOPE OF THE PROJECT

Current Scope: Validating student registration using ZPRN mappings, managing resource catalogs (Rent/Exchange), lost-and-found report listings, and local database transaction routing.

Future Scope:

- Live chat messaging between borrowers and owners.
- Push notifications for overdue items.
- Multi-department integration allowing other campus branches (Mechanical, Civil, IT) to participate.
- Secure integration of UPI payment APIs for deposit handling.

14. REFERENCES

1. *Nodemailer Documentation*, "SMTP Transport configuration guide," <https://nodemailer.com/smtp/>
2. *Tailwind CSS Documentation*, "Utility-First Fundamentals & Responsive Design," <https://tailwindcss.com/docs/>
3. *ExpressJS Reference Guide*, "Routing and Middleware Architecture," <https://expressjs.com/>
4. *Visual Studio Code Reference*, "Managing projects and source control with Git," <https://code.visualstudio.com/docs>

(Name Of Guide)
Project Guide

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